Introduction of IIR filter

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Two categories of discrete-time filters

Digital filters are divided into two categories according to their impulse response:

Infinite Impulse Response filters (IIR filters)

$$y(n) = \sum_{i=0}^{N} a_i\, x(n-i) - \sum_{i=1}^{N} b_i\, y(n-i)$$

$$H(z) = \frac{\sum_{i=0}^{N} a_i\, z^{-i}}{1 + \sum_{i=1}^{N} b_i\, z^{-i}}$$

Finite Impulse Response filters (FIR filters)

$$y(n) = \sum_{i=0}^{N} a_i\, x(n-i)$$

$$H(z) = \sum_{i=0}^{N} a_i\, z^{-i}$$

→ An FIR filter has 5 to 10 times larger realization structure than a corresponding IIR filter.
→ An FIR filter is always stable.
→ An FIR filter is called a **non-recursive structure**, while an IIR filter is called a **recursive structure**.
→ An FIR filter is **less sensitive** to coefficient changes and rounding errors than an IIR filter.

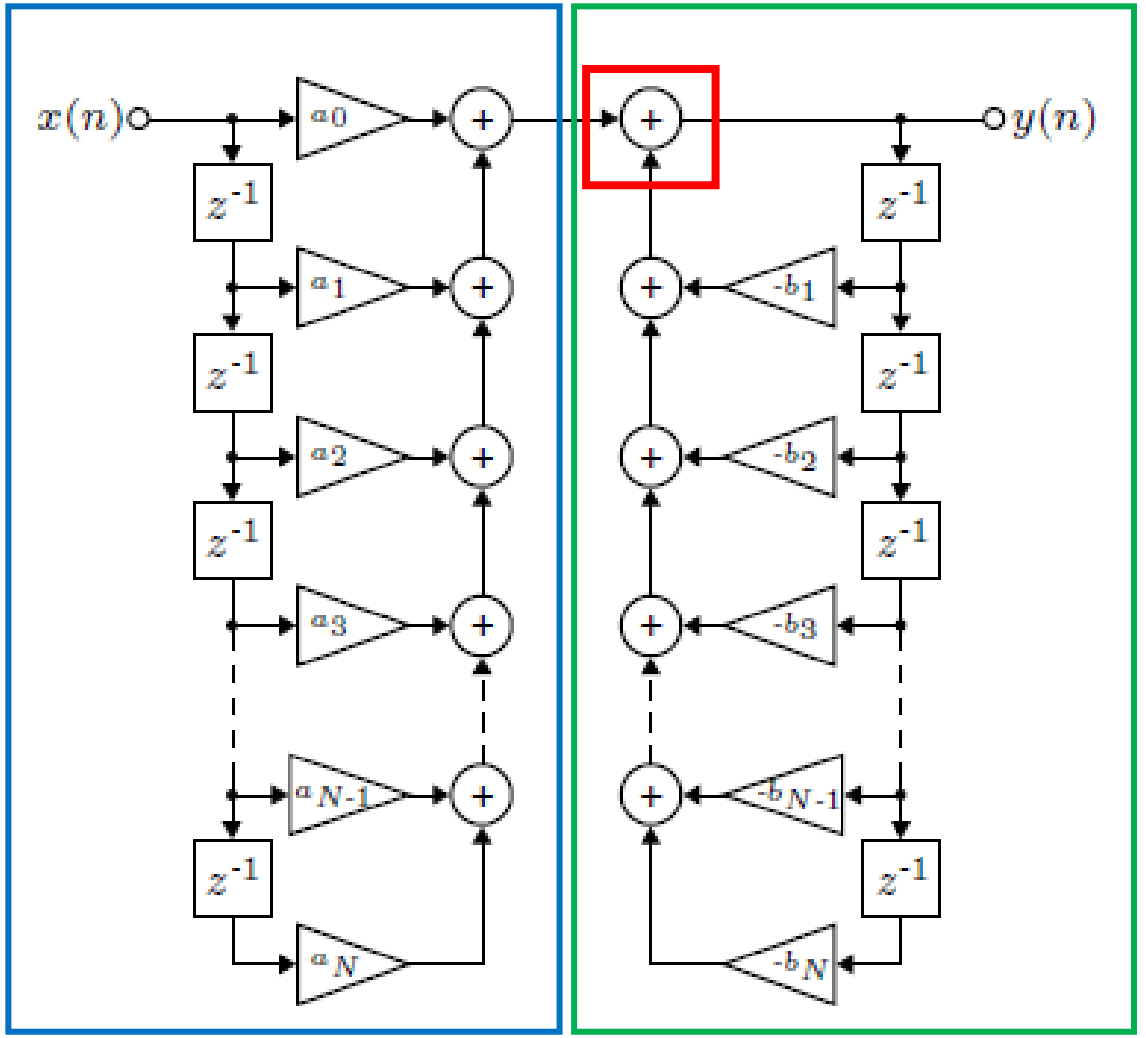
Recap: realization

Transfer function

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^{N} a_i \, z^{-i}}{1 + \sum_{i=1}^{N} b_i \, z^{-i}}$$

Difference equation

$$y(n) = \sum_{i=0}^{N} a_i x(n-i) - \sum_{i=1}^{N} b_i y(n-i)$$



Design of digital IIR filter

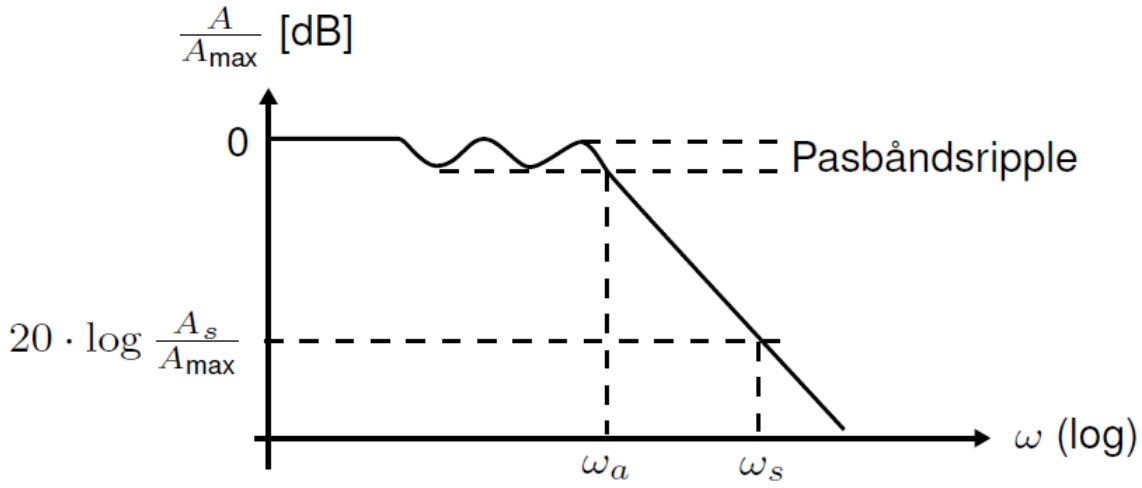
An IIR filter is designed by following the procedure

1. The filter's specifications are drawn up (analog filter)
2. Convert the analog filter to digital filter: z-domain transfer function is made
3. Choose a realization structure
4. Implement the design through program or hardware

The following methods will be covered in this course:
→ **Matched z-transform**
→ **Impulse invariant z-transform**
→ Bilinear z-transform (more popular, next lecture)

Recap - Filter specification

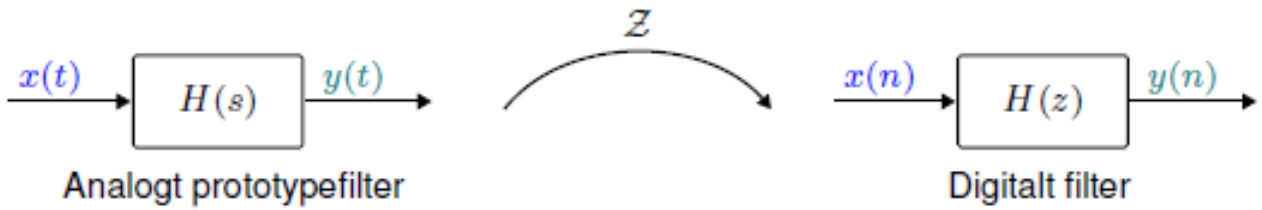
1. A **cutoff frequency ω_a** which indicates the upper limit of the passband.
2. A **stopband frequency ω_s** at which a **stopband attenuation A_s** is specified.
3. Information of passband ripple.



Transformation from an analog prototype filter

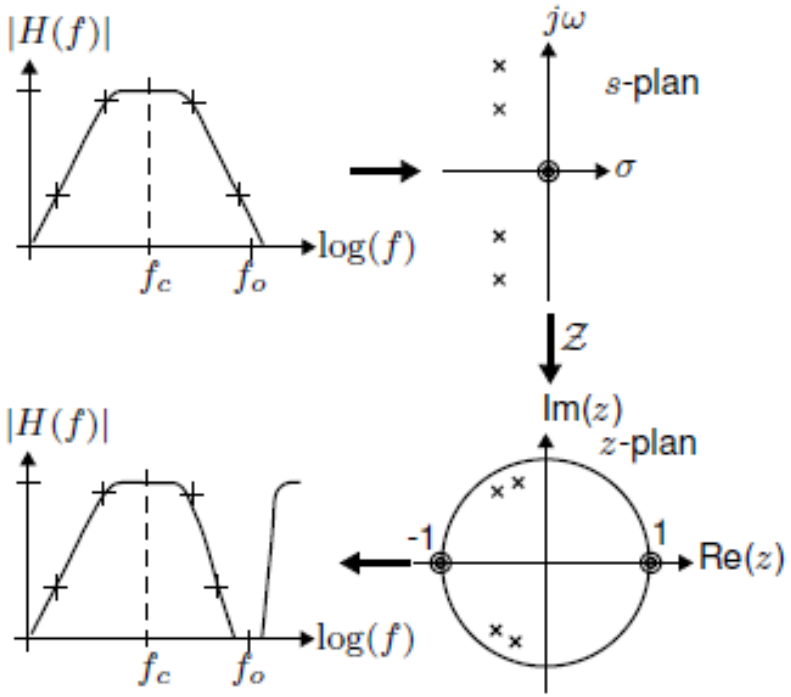
$$H(s) = \frac{\sum_{i=0}^{M} A_i s^i}{\sum_{i=0}^{N} B_i\, s^i}$$

$$H(z) = \frac{\sum_{i=0}^{M} a_i z^{-i}}{\sum_{i=0}^{N} b_i\, z^{-i}}$$



Matched z-transform

The transformation from s-domain to z-domain:
Transfer the **poles** and **zeros** from the s-plane to the z-plane. This will give a similar filter response.



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Matched z-transform

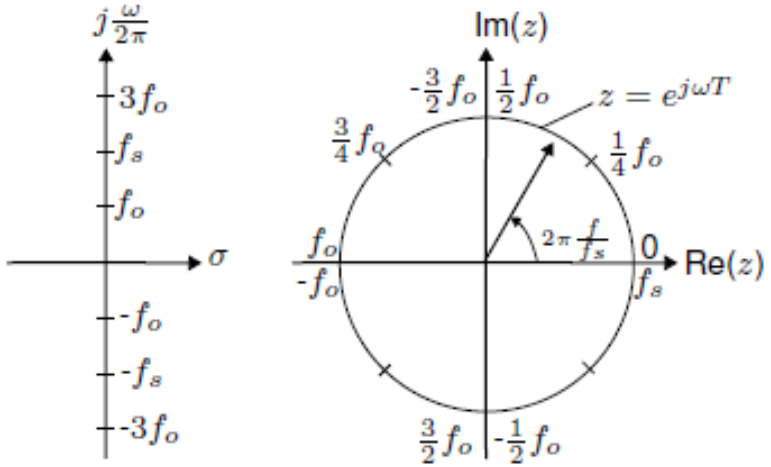
With matched z-transform, the **poles and zeros** of the prototype filter are transferred directly to the z-domain using the following

$$X_s(s) = X(z) \quad \text{when} \quad z = e^{sT}$$

Same equation used previously to relate the z-plane and the s-plane.

Laplace transform of **sequence** $x(n)$

$$X_s(s) = \sum_{n=0}^{\infty} x(n)e^{-snT}$$



z transform of **sequence** $x(n)$

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

s domain

z domain

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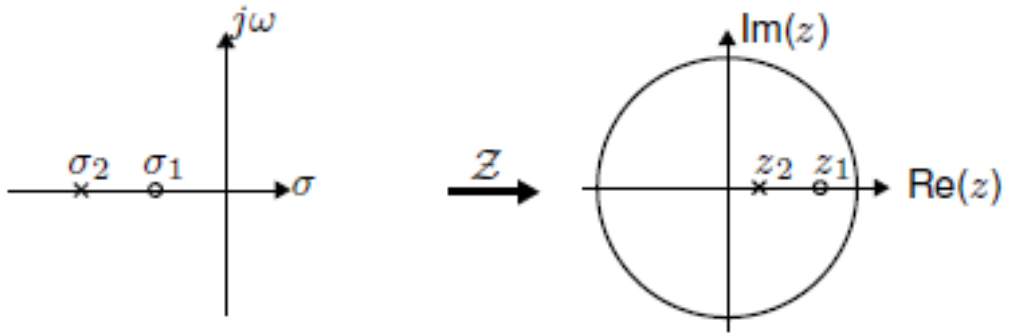
Matched z-transform for 1st order system

A 1st order transfer function with 1 zero is given by

$$H(s) = \frac{s + A_0}{s + B_0} = \frac{s - \sigma_1}{s - \sigma_2}$$

Where $\sigma_1 = -A_0$ is a real **zero** and $\sigma_2 = -B_0$ is a real **pole**

When using matched z-transformation, the **zero** and the **pole** can be calculated as
$z_1 = e^{s_1 T} = e^{\sigma_1 T}$ and $z_2 = e^{s_2 T} = e^{\sigma_2 T}$



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Matched z-transform for 1st order system

A digital first-order filter based on matched z-transform has the form

$$H(z) = \frac{z - e^{\sigma_1 T}}{z - e^{\sigma_2 T}}$$

$$H(z) = \frac{1 - e^{\sigma_1 T} z^{-1}}{1 - e^{\sigma_2 T} z^{-1}}$$

Matched z-transform for 1st order system

Let's consider a 1st order system with no zero

$$H(s) = \frac{\sigma_1}{s - \sigma_1}$$

The above has only 1 pole, and it can be converted to the following

$$H(z) = \frac{1}{1 - e^{\sigma_1 T} z^{-1}}$$

where σ_1 is the pole of $H(s)$ and T is the sampling interval.

There is a problem above:
Try to substitute $z = 1$

$$H(1) = \frac{1}{1 - e^{\sigma_1 T}} \neq 1$$

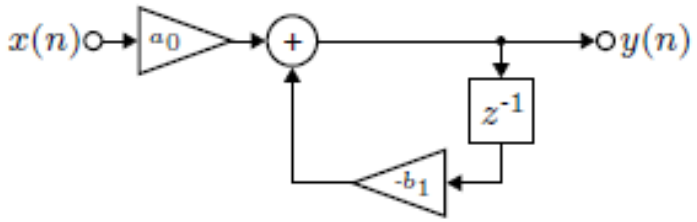
What does $z = 1$ mean?
$z = 1\, e^{j\omega T}$ where $\omega = 0$
It represents the DC gain!
It corresponds to the origin of the s-plane.
$H(s = 0) = 1$.

Matched z-transform for 1st order system

To solve this problem, we introduce an **amplification factor**

$$H(z) = \frac{1 - e^{\sigma_1 T}}{1 - e^{\sigma_1 T} z^{-1}}$$

A digital realization structure for the first order system is shown here.



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Example

Given an analog 1st order lowpass filter

$$H(s) = \frac{\omega_a}{s + \omega_a} = \frac{1885}{s + 1885}$$

Whose cutoff frequency $\omega_a = 2\pi f_a$ and $f_a = 300\text{Hz}$.

It has **1 pole**: $s_1 = \sigma_1 = -1885$

We want to determine a digital filter by **matched z-transform** with a sampling frequency of f_s =16 kHz.
The coefficients of the digital filter become

The original s-transfer function has no zeros, so we use the following

$$H(z) = \frac{1 - e^{\sigma_1 T}}{1 - e^{\sigma_1 T} z^{-1}}$$

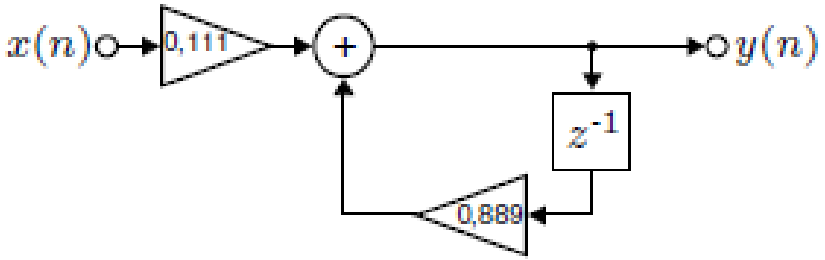
$$1 - e^{\sigma_1 T} = 0.1111$$
$$e^{\sigma_1 T} = 0.8889$$

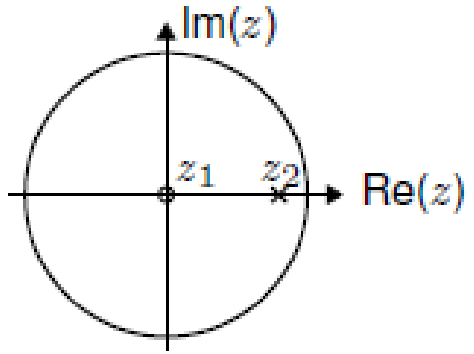
The digital low-pass filter transfer function is

$$H(z) = \frac{0.1111}{1 - 0.8889\, z^{-1}}$$

Example

The digital filter $H(z)$ has zero $z = 0$ and pole $z = 0.8889$
The direct type II realization structure is shown below.

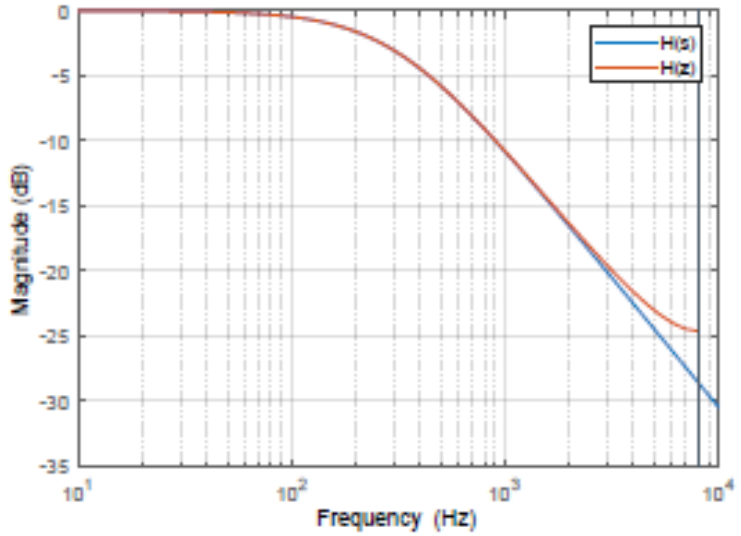




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Example

When comparing the amplitude characteristics of the two filters, you can find that they differ at high frequencies (close to the **Nyquist frequency** f_o).



Due to this inherent limitation, the matched z-transform method normally is used when $f_a \ll f_o$

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Matched z-transform for higher order system

For a higher order system with complex conjugate pole and zero pair, the transfer function can be written after factorization

$$H(s) = \frac{(s - z_1)(s - z_1^*)}{(s - p_1)(s - p_1^*)} \cdot \frac{(s - z_2)(s - z_2^*)}{(s - p_2)(s - p_2^*)} \cdot \ldots \cdot \frac{(s - z_N)}{(s - p_N)}$$

By converting 's' -> 'z', and replacing s_i with $z = e^{sT}$

$$H(z) = \frac{zeros}{(z - e^{p_1 T})(z - e^{p_1^* T})} \cdot \frac{zeros}{(z - e^{p_2 T})(z - e^{p_2^* T})} \cdot \ldots \cdot \frac{zero}{z - e^{p_N T}}$$

For zeros, you have the flexibility to convert some of them, to maintain the shape of H(s).
You are allowed to omit some zeros to simplify the transfer function H(z)

Determine the zeros for $H(z)$ so that $H(z = 1) = 1$

Normally, a **cascade structure** is used for the realization when matched z transform method is used.

Design procedure (matched z-transform)

1. Determine the analog filter's transfer function $H(s)$.
2. Determine the analog filter’s poles and zeros.
3. Convert the **poles** in s-domain to z-domain
4. Determine the coefficients of the digital transfer function. The numerator may be modified to have $H(z = 1) = 1$
5. Realize the transfer function as a **cascade structure**.

Example: 2nd order system

For a 2nd order system with complex conjugate pole and zero pair, the transfer function can be written

$$H(s) = \frac{s^2 + A_1 s + A_0}{s^2 + B_1 s + B_0}$$

Which have **zeros** $s_1 = \sigma_1 + j\omega_1$, $s_1^* = \sigma_1 - j\omega_1$, and **poles** $s_2 = \sigma_2 + j\omega_2$, $s_2^* = \sigma_2 - j\omega_2$

$$H(s) = \frac{(s - s_1)(s - s_1^*)}{(s - s_2)(s - s_2^*)}$$

By converting 's' -> 'z', and replacing s_1, s_2 with $z = e^{sT}$

$$H(z) = \frac{(z - e^{\sigma_1 T} e^{j\omega_1 T})(z - e^{\sigma_1 T} e^{-j\omega_1 T})}{(z - e^{\sigma_2 T} e^{j\omega_2 T})(z - e^{\sigma_2 T} e^{-j\omega_2 T})}$$

Using Euler formula $\cos(x) = (e^{jx} + e^{-jx})/2$, we have

$$H(z) = \frac{z^2 - (2e^{\sigma_1 T}\cos(\omega_1 T))z + e^{2\sigma_1 T}}{z^2 - (2e^{\sigma_2 T}\cos(\omega_2 T))z + e^{2\sigma_2 T}} = \frac{1 - (2e^{\sigma_1 T}\cos(\omega_1 T))z^{-1} + e^{2\sigma_1 T}z^{-2}}{1 - (2e^{\sigma_2 T}\cos(\omega_2 T))z^{-1} + e^{2\sigma_2 T}z^{-2}}$$

Example: 2nd order system

A 2nd order low-pass filter is given as

$$H(z) = \frac{a_0}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

From the previous slide we know

$$H(z) = \frac{1 - (2e^{\sigma_1 T}\cos(\omega_1 T))z^{-1} + e^{2\sigma_1 T}z^{-2}}{1 - (2e^{\sigma_2 T}\cos(\omega_2 T))z^{-1} + e^{2\sigma_2 T}z^{-2}}$$

Here, I omit all the zeros!
So a_0 is a constant.

In this case,

$$b_1 = -2e^{\sigma_2 T}\cos(\omega_2 T) \quad \text{and} \quad b_2 = e^{2\sigma_2 T}$$

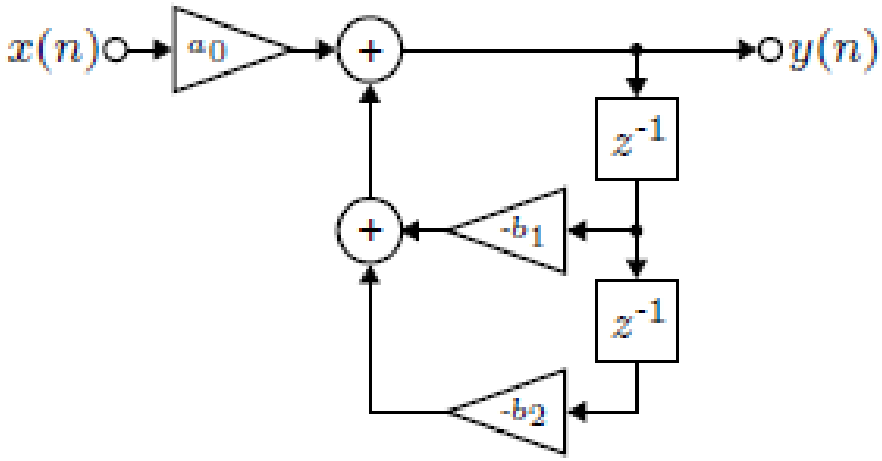
If the **DC gain** for the low-pass filter is to be 0 dB then it applies that

$$a_0 = 1 + b_1 + b_2$$

We can also change the zeros positions during the conversion, but we need to assure $H(z = 1) = 1$

Example: 2nd order system

The realization structure for the low-pass filter is given as shown below.



Exercise

Design a digital 2nd order Butterworth low pass filter with cutoff frequency $f_a = 800$Hz using Matched z-transform with sampling rate of 8 kHz.

A **frequency-normalized** transfer function for a 2nd order Butterworth low-pass filter is

$$H(s) = \frac{1}{s^2 + \sqrt{2}\, s + 1}$$

It has two poles $s = -0.7071 \pm j0.7071$

Denormalized poles with cutoff frequency $f_a = 800$ Hz are
$s = -3554.5 \pm j3554.5$

Frequency normalize:
$s \rightarrow s/\omega_a$
(recap in lecture 'Filters')

The filter coefficients can now be calculated using the formulas
$b_1 = -2e^{\sigma_2 T}\cos(\omega_2 T)$ and $b_2 = e^{2\sigma_2 T}$ and $a_0 = 1 + b_1 + b_2$

Can you calculate the final discrete-time transfer function**?**

$$H(z) = \frac{a_0}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

If you run Matlab:
[b,a] = butter(2,2*pi*fa,'s');
H_s = tf(b,a)

You can have the denormalized transfer function:

$$H(s) = \frac{2.527e07}{s^2 + 7109\, s + 2.527e07}$$

Then you can use the following function to find zeros and poles

- roots()
- pzmap()

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Matlab function: Hz=c2d(Hs, Ts, method)

→ Create the required continuous-time transfer function with required specs: Hs
→ Determine the sampling period Ts
→ Method can be
  → **'matched' — Zero-pole matching method**
  → 'impulse' — Impulse invariant discretization
  → 'tustin' — Bilinear (Tustin) method.

$H(z = 1) = 1$ applies as well!

```
fa = 800;
[b,a] = butter(2,2*pi*fa,'s');
H_s = tf(b,a)

fs = 8000;
H_z = c2d(H_s, 1/fs, 'matched')

>> H_z =
   0.1266 z + 0.1266
  ----------------------
  z^2 - 1.158 z + 0.4112
```

Example

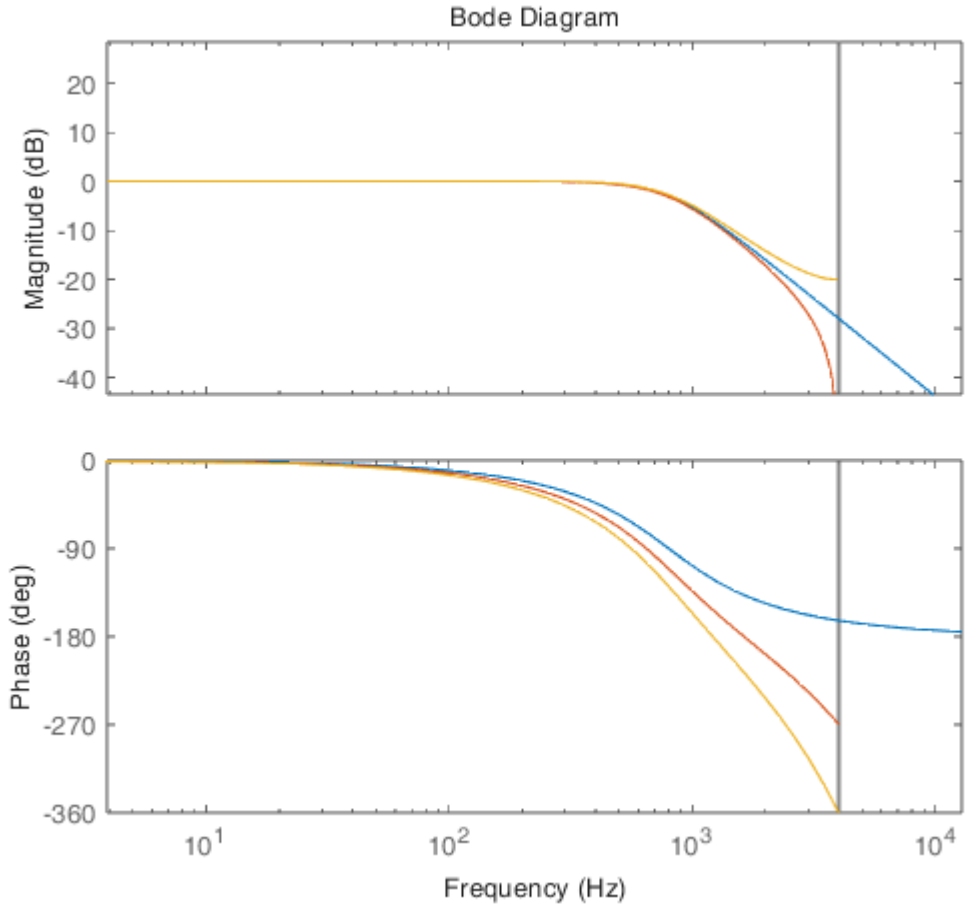
You can see that the digital filter is a very aliased version of the original analog filter $H(s) = \frac{2.527e07}{s^2 + 7109\,s + 2.527e07}$ (blue).

Matlab calculated (red)

$$H_z = 0.1266\,\frac{z+1}{z^2 - 1.158\,z + 0.4112}$$

Manual calculated (yellow)

$$H_z = \frac{0.2532}{z^2 - 1.158\,z + 0.4112}$$



Getting worse when f closer to f_0

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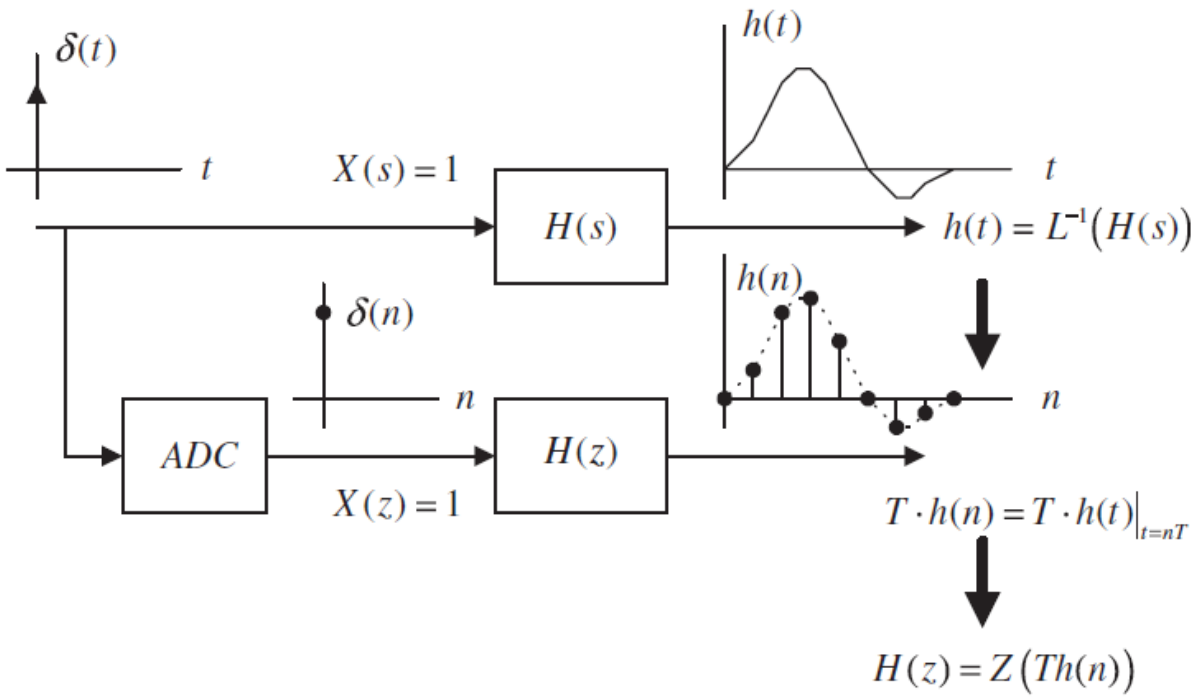
Impulse invariant z-transform

The analog impulse response can be achieved by using inverse Laplace transform of analog filter $H(s)$.

$$h(t) = \mathcal{L}^{-1}[H(s)]$$

Then we sample this analog impulse response with a sampling interval of T. Also, T is used as a scale factor.

$$T \cdot h(n) = T \cdot h(nT), \quad n \geq 0$$



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Impulse invariant z-transform

Given a Nth order filter

$$H(s) = \frac{\sum_{i=0}^{M} A_i s^i}{\sum_{i=0}^{N} B_i s^i} \qquad M \leq N$$

Using partial fraction solution

$$H(s) = \sum_{i=1}^{N} \frac{r_i}{s - p_i}$$

Matlab function:
residue()

where p_i is the analog filter's pole

Impulse invariant z-transform

The transfer function

$$H(s) = \sum_{i=1}^{N} \frac{r_i}{s - p_i}$$

Using inverse Laplace transformed

$$h(t) = \mathcal{L}^{-1}[H(s)] = \sum_{i=1}^{N} r_i e^{p_i t}$$

The impulse response of the digital filter thus becomes

$$h(n) = h(nT) = \sum_{i=1}^{N} r_i e^{p_i nT}$$

By z-transform of $h(n)$

$$H(z) = T \sum_{i=1}^{N} r_i \frac{z}{z - e^{p_i T}} = T \sum_{i=1}^{N} \frac{r_i}{1 - e^{p_i T} z^{-1}}$$

ZT5	$e^{s_0 nT}$	$\frac{z}{z - e^{s_0 T}}$

Design procedure (Impulse invariant z-transform)

1. Determine the analog filter's transfer function $H(s)$.
2. Conduct partial fraction decomposition $H(s) = \sum_{i=1}^{N} \frac{r_i}{s-p}$.
3. Convert each section from s-domain to z-domain $H(z) = T \sum_{i=1}^{N} \frac{r_i}{1-e^{p_i T} z^{-1}}$
4. Realize the transfer function as a **parallel structure**.

Example

A transfer function for a 2nd order Butterworth low-pass filter with cutoff frequency $f_a = 800$ Hz

$$H(s) = \frac{2.527e07}{s^2 + 7109\,s + 2.527e07}$$

It has two poles $s = -3554.5 \pm 3554.6i$

Using partial fraction decomposition (Matlab residue()), $H(s)$ can be written

$$H(s) = \sum_{i=1}^{N} \frac{r_i}{s - p_i} = -\frac{3554.5j}{s + 3554.5 - 3554.6j} + \frac{3554.5j}{s + 3554.5 + 3554.6j}$$

According to

$$H(z) = T \sum_{i=1}^{N} \frac{r_i}{1 - e^{p_i T} z^{-1}}$$

Sampling interval $T = 1/8k$

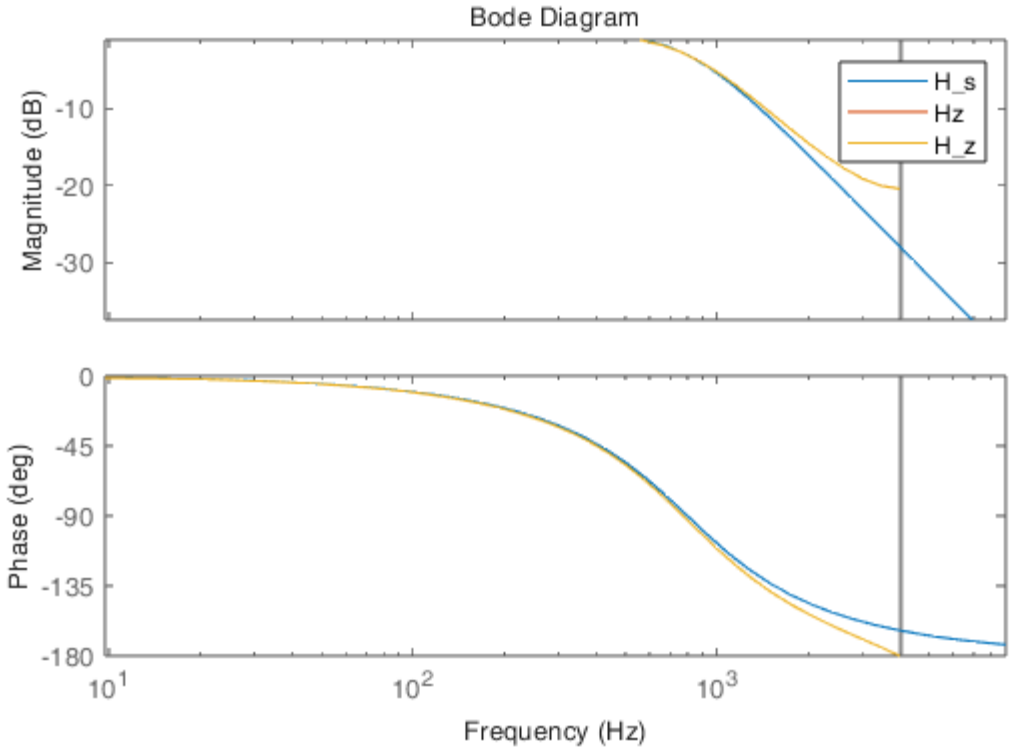
$$H(z) = \frac{-0.4443j}{1 - (0.579 + 0.2757j)z^{-1}} + \frac{0.4443j}{1 - (0.579 - 0.2757j)z^{-1}}$$

Matlab

```
fa = 800;
[b,a] = butter(2,2*pi*fa,'s');
H_s = tf(b,a)

fs = 8000;
H_z = c2d(H_s, 1/fs, ‘impulse’)

>> H_z =
        0.2449 z
  ----------------------
  z^2 - 1.158 z + 0.4112
```



Blue: H(s)
Red: H(z) using equation
Yellow: H(z) using Matlab c2d()

Exercise

Use Impulse invariant z transform method to derive a 4th order Butterworth bandpass filter, cutoff frequency $f_{a1} = 300Hz$, $f_{a2} = 800Hz$, sampling frequency $f_s = 8000Hz$.

1. Find the analog filter transfer function
   → [b,a] = butter(4,[2*pi*300, 2*pi*800],'bandpass',’s’)
   → H_s = tf(b, a)
2. Partial fraction decomposition
   → [r,p,k]= residue(b,a);
3. Using $H(z) = T\sum_{i=1}^{N} \frac{r_i}{1-e^{p_i T} z^{-1}}$, find the discrete-time transfer function
4. Do a bode() plot to check if all the specs are satisfied
5. Try to draw the block diagram realization

(option) Try using function c2d() in place of step 2 to 3, and compare results.